

Chilled Water Plant Equipment Schedule & Piping Schematic

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Project: Lakefront Office Tower — Phase II

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SECTION 23 64 16 — CENTRIFUGAL WATER CHILLERS

1.0 SCOPE OF WORK

Furnish and install centrifugal water chillers, cooling towers, chilled water pumps, condenser water pumps, and associated piping, valves, and controls per drawings M3.01–M3.06. System provides N+1 chiller redundancy.

2.0 CHILLED WATER PLANT PIPING SCHEMATIC

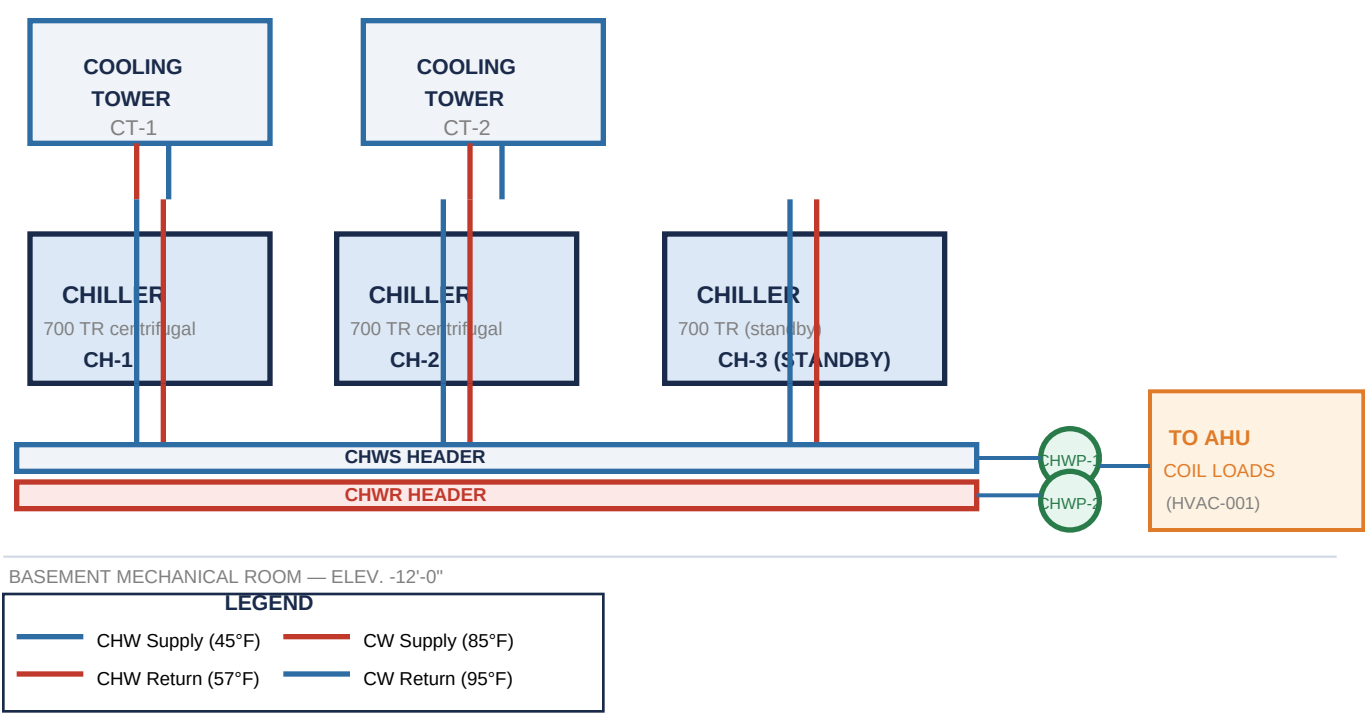


Figure 4 — Chilled Water Plant Piping Schematic (Not to Scale)

3.0 CHILLER SCHEDULE

Tag	Capacity (Tons)	Type	Refrigerant	IPLV (kW/Ton)	COP	Notes
CH-1	700	Centrifugal	R-134a	0.480	6.5	Primary
CH-2	700	Centrifugal	R-134a	0.480	6.5	Primary
CH-3	700	Centrifugal	R-134a	0.480	6.5	N+1 Standby

4.0 COOLING TOWER SCHEDULE

Tag	Capacity (Tons)	Flow (GPM)	Range (°F)	Fan (HP)	Approach (°F)	Type
CT-1	875	2,100	10	100	7	Induced draft, crossflow

Tag	Capacity (Tons)	Flow (GPM)	Range (°F)	Fan (HP)	Approach (°F)	Type
CT-2	875	2,100	10	100	7	Induced draft, crossflow

5.0 PUMP SCHEDULE

Tag	Service	Flow (GPM)	Head (ft)	Motor (HP)	Drive	Notes
CHWP-1	CHW Primary	2,800	80	75	VFD	Lead
CHWP-2	CHW Primary	2,800	80	75	VFD	Standby
CHWP-3	CHW Secondary	3,500	60	75	VFD	Lead
CHWP-4	CHW Secondary	3,500	60	75	VFD	Standby
CWP-1	Condenser	4,200	60	100	VFD	Lead
CWP-2	Condenser	4,200	60	100	VFD	Standby

END OF SECTION 23 64 16